Lab - Research a Hardware Upgrade

Introduction

Use the Internet, a newspaper, or a local store to gather information about hardware components. Your customer's computer currently has one module of 2 GB of RAM, a 500 GB hard disk drive, and a PCIe video adapter card with 256 MB of RAM. Your customer wants to be able to play advanced video games.

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Instructions

Step 1: Research memory options.

Shop around, and in the table below list the brand, model number, features, and cost for five different 8 GB modules of DDR3.

Brand and Model Number	Features	Cost
Kingston HyperX FURY 8GB DDR3 Desktop RAM	*Features **automatic overclocking** (Plug-and-Play up to 1866MHz), a stylish low-profile asymmetric aluminum heat spreader, and strict 1.5V low-power delivery optimized for stable everyday workloads.*	*Ranges to* *₱1,076-₱1,736*
CORSAIR Vengeance 8GB DDR3 Desktop RAM	*Built heavily around an **8-layer PCB design with factory-screened ICs** for massive manual overclocking overhead. It incorporates a taller, aggressive aluminum cooling armor and supports Intel XMP 1.3 profiles.*	*Ranges to* *₱1,184 to ₱1,250*
A-Tech 8GB DDR3 1600MHz Desktop Memory	*A reliable **value-driven standard module** strictly meeting JEDEC PC3-12800 guidelines. It features a bare non-ECC layout built to easily maximize capacity on older budget legacy setups without throwing BIOS faults.*	*Ranges to* *₱1,300 to ₱1,500*
Team Group Elite 8GB DDR3 1600MHz Desktop Ram	*Features a high-quality aluminum **heat spreader** for thermal control. It operates at a standard 1.5V and fully adheres to rigid JEDEC standards, ensuring seamless, error-free integration into consumer desktop builds.*	*Ranges to* *788 to ₱1,443*
Patriot Signature Line 8GB DDR3 1600Mhz Desktop Ram	*Designed entirely as a **low-cost "value" RAM stick** without a bulky heat shield. This makes it perfect for compact, tightly packed desktop chassis. It features a dual-sided design with factory-tested ICs that prioritize long-term system stability over high overclocking speeds.*	*Ranges to* *₱750 to ₱1,100*

Question:

Based on your research, which RAM would you select? Be prepared to discuss your decisions regarding the RAM you select.

In choosing a ram for advanced video games, I would prefer/recommend Kingston HyperX FURY 8GB DDR3 Desktop RAM as in its feature it can overclock from 1600Mhz to 1866Mhz, have an asymmetric heat spreader and a strict 1.5V low-power delivery, which makes gaming with this RAM much smoother than the rest, but It would be much preferable to use dual channel for a ram (2 x 8) *I* **(**2 x 4) to split the memory and to not

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overload **or congest** a single ram stick **when doing heavy workloads and gaming**, even if it has overclocking.

Step 2: Research hard disk drive options.

Shop around, and in the table below list the brand, model number, features, and cost for a 3 TB 7200 rpm SATA 3 hard disk drive and a 500GB or 1TB solid state drive (SSD).

Brand and Model Number	Features	Cost
Western Digital Blue 3.5-inch Internal HDD 3TB (WD30EZRZ)	*3.5-inch desktop drive, 64MB cache, optimized for everyday computing reliability.*	***₱10,999.00 (On Lazada PH)***
Western Digital Blue SA510 SATA 2.5-inch SSD 1TB	*Up to 560MB/s sequential read, lacks mechanical parts, great for upgrading older systems.*	***₱15,000+ (On uBuy.com.ph)***
Toshiba P300 High-Performance Desktop HDD 3TB (HDWD130)	*Dual-stage actuator for accurate head placement, built-in internal shock sensors.*	***₱12,000+ (On eBay)***
Crucial MX500 SATA 2.5-inch SSD 1TB	*Integrated Power Loss Immunity, hardware-based AES 256-bit encryption.*	***₱9,000+ (On bermorzone.com.ph)***
Samsung 870 EVO SATA III 2.5-inch SSD 500GB	*Intelligent TurboWrite technology, top-tier reliability, supported by Samsung Magician software*	***₱12,000+ (On DataBlitz.com.ph)***

Question:

Based on your research, which hard disk drive would you select? Be prepared to discuss your decisions regarding the hard disk drive you select.

*The best choice in all of this hard disk would be both of the **Western Digital Blue hard disk's**, as both hard drive's run on the same ecosystem, if you would be using both an SSD and HDD, due to their features in balancing performance while still being able to save some of your budget since both are cheaper than **Toshiba P300 Performance Desktop HDD 3TB** and **Samsungs 870 EVO SATA III 2.5 Inch SSD 500GB**, but If I were only going to choose only 1 hard drive I would recommend **Crucial MX500 SATA 2.5-inch SSD 1TB*** if you would prioritize data safety and a fast performance still capable for gaming for a single ROM while also having a massive storage of 1TB and saving some budgets since it is cheaper than the others.

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Step 3: Research video adapter card options.

Shop around, and in the table below list the brand, model number, features, and cost for five different PCIe video adapter cards with at least 1 GB of RAM.

Brand and Model Number	Features	Cost
NVIDIA GeForce GT 730	***Features a Memory Capacity*** *of 1 GB to 4 GB (DDR3 or GDDR5)* ***Bus Interface:*** *PCIe 2.0 x8 / x16* *provides multi-monitor configurations for up to three displays, standard low-profile formats for compact systems, optional passive cooling for completely silent operation, and dedicated hardware decoding to handle smooth 1080p video streaming.*	***Ranges to*** ***₱4,700 - ₱7,000+***
AMD Radeon R5 230	***Memory Capacity:*** *1 GB to 2 GB (DDR3)* ***Bus Interface:*** *PCIe 2.0 / 2.1 x16* *offers legacy video outputs like native VGA, DVI, and HDMI to connect older monitors, relies on an ultra-low power consumption rating of under 20 Watts, fits into tight spaces via a single-slot footprint, and serves as a basic display adapter for processors without integrated graphics.*	***Ranges to*** ***₱1,500- ₱4,800+***
NVIDIA GeForce GT 1030	***Memory Capacity:*** *2 GB (GDDR4 or GDDR5)* ***Bus Interface:*** *PCIe 3.0 x4* *utilizes the efficient 14nm Pascal architecture, decodes modern video formats like HEVC and VP9 for 4K media playback at 60Hz via HDMI 2.0, processes light esports gaming at 1080p, and operates over a PCIe 3.0 x4 bus design to reduce power consumption.*	***Ranges to*** ***₱4,500 – ₱6,200***
AMD Radeon RX 550	***Memory Capacity:*** *2 GB to 4 GB (GDDR5)* ***Bus Interface:*** *PCIe 3.0 x8* *delivers full hardware compliance with DirectX 12 and Vulkan APIs to run newer software, features AMD FreeSync compatibility to eliminate on-screen tearing, utilizes Frame Rate Target Control to restrict unnecessary power and heat generation, and grants access to the modern Radeon Software suite.*	***Ranges to*** ***₱6,000 – ₱9,000***
NVIDIA GeForce GTX 750 Ti	***Memory Capacity:*** *1 GB to 2 GB (GDDR5)* ***Bus Interface:*** *PCIe 3.0 x16* *runs entirely off motherboard slot power using highly efficient Maxwell architecture, integrates a wider 128-bit memory bus to transfer heavy game textures quickly, includes hardware-accelerated NVIDIA ShadowPlay to record gameplay with minimal performance loss, and supports smooth display synchronization via G-Sync technology.*	***Ranges to*** ***₱5,500 – ₱6,500***

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Question:

Based on your research, which video adapter card would you select? Be prepared to discuss your decisions regarding the video adapter card you select.

*The best option overall if the hardware or motherboard only supports ddr3 is **NVIDIA GeForce GT 1030,***

*while **AMD Radeon RX 550 and NVIDIA GeForce GTX 750 Ti** have better specs, we need to consider not to hardware or driver compatibility issues that would make the performance to depreciate from its intended performance with hardware it is compatible with, and with **GT 1030 you can still play in 4k video playback** and **1080p esports gaming** and the performance you are getting is what you paid for, you can also still use the video adapter/GPU when upgrading to a **DDR4 system and DDR5(In some cases with a Standard PCIe Slot)** since it is still modern enough to receive stable driver support.*

End of Document

Sources:

RAM (Random Access Memory):

Kingston HyperX FURY 8GB DDR3 Desktop RAM: https://forums.tomshardware.com/threads/best-ddr3-ram-for-gaming.2888878/, https://www.kingston.com/en/memory,

CORSAIR Vengeance 8GB DDR3 Desktop RAM: Google's AI Mode, Lazada PH, Shopee PH.

A-Tech 8GB DDR3 1600MHz Desktop Memory: https://atechmemory.com/products/a-tech-8gb-1x8gb-pc3-12800u-ddr3-1600-dimm-desktop-ram-memory-upgrade-module?variant=4,

Team Group Elite 8GB DDR3 1600MHz Desktop Ram: https://easypc.com.ph/products/team-elite-8gb-1x8-memory-1600mhz-ddr3-with-heatspreader?variant=39928083677355,

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HARD DRIVES/HARD DRIVES/ROM(Read-Only Memory):

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Western Digital Blue SA510 SATA 2.5-inch SSD 1TB: https://www.ubuy.com.ph/productde/TSTOCH21E-wd-blue-sa510-2-5-ssd-1tb-wds100t3b0a?srsltid=AfmB

Toshiba P300 High-Performance Desktop HDD 3TB (HDWD130): https://www.ebay.com/itm/233260319600?chn=ps&mkevt=1&mkcid=28&google_free_listing_action=view_item&srsltid=AfmBOor8rVG6q3KyDtxRQNgNFiVo9lA9Yg6mu1GpdMPUemidpwe8cFw73Nk

Crucial MX500 SATA 2.5-inch SSD 1TB: https://bermorzone.com.ph/shop/storage-devices/solid-state-drives/crucial-mx500-1tb-3d-nand-sata-2-5-inch-internal-ssd/?srsltid=AfmBOooGOnEVGviKFLT4cMudlbiip-D4YEdP5kBNCYGLaOFmhmKzgiVeFrY

Samsung 870 EVO SATA lll 2.5-inch SSD 500GB: https://ecommerce.datablitz.com.ph/products/samsung-870-evo-500gb-sata-iii-2-5-ssd-mz-77e500bw

PCIe video adapter cards with at least 1 GB of RAM/ Video Graphics Card/ GPU:

***NVIDIA GeForce GT 730:** https://www.ubuy.com.ph/product/3UZ20FYMA-asus-geforce-gt-730-2gb-gddr5-low-profile-graphics-card-for-silent-htpc-builds-with-i-o-port-brackets?srsltid=AfmBOoqNE-M3QQ7aCVv8MM6tv9yHowoJWKJXMrPUneD4yFJmB5CTP8lOIMk, https://www.desertcart.ae/products/725971265-asustek-nvidia-geforce-gt-730-2gb-gddr5-hdmi-2-0b*

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AMD Radeon R5 230: ***https://www.lazada.com.ph/products/soyo-soyo-amd-radeon-r5-230-2gb-ddr3-64bit-graphics-card-pcie20x16-hdmi-vgadvi-video-card-ati-40nm-lowprofile-card-desktop-computer-i15520591486-s133241341750.html?from_gmc=1&fl_tag=1,***

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AMD Radeon RX 550: ***https://www.ubuy.com.ph/product/F4YIlJZA0-powercolor-karta-graf-rx-550-2gb-gddr5?srsltid=AfmBOogLo6-Id8E7NMUzf-5O3zXTJY3hdafLyjttjbw9DpUhbJrpE1XlpNI***

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